

# NATIONAL COLLEGE OF IRELAND: SCHOOL OF COMPUTING

Master of Science in Data Analytics (MSCDAD) | Research Practicum

## WEEKLY PROGRESS REPORT: WEEK 1

|                          |                                                                                                          |
|--------------------------|----------------------------------------------------------------------------------------------------------|
| Student Name:            | Rohitkumar Amrittal Jaiswal (ID: x25119613)                                                              |
| Supervisor / Lecturer:   | Dr. Thanos Staikopoulos                                                                                  |
| Project Title:           | Coupling Port Congestion & Customs Delay Prediction with Carbon-Aware Multi-Objective Route Optimisation |
| Reporting Period / Date: | Week 1 (Kickoff, Technical Initialization & Baseline Benchmark)   Tuesday, 29 September 2026             |

### 1. Activities Completed This Week (What?)

- **Project Proposal & Ethics Sign-Off:** Finalised the 4,691-word Research in Computing Proposal PDF (CA2) streamlining scope into 3 coupled modules (C1, C2, C3). Curated 7 peer-reviewed reference papers and signed the official Ethics Consideration Form.
- **Repository Architecture & Packaging:** Initialized the formal Git repository with modern PEP 621 packaging (irishlogix v0.1.0) and automated testing harness.
- **Automated CSO Data Pipeline (cso\_loader.py):** Integrated the Central Statistics Office (CSO) PxStat API; successfully ingested and parsed 444 vessel arrivals (TBQ01), 2,331 Ro-Ro freight records (TBQ04), and 3,600 trade region records (TBQ05). Created unified port congestion matrix.
- **Jupyter Notebook Development:** Created and executed notebooks/01\_week1\_data\_ingestion\_and\_exploration.ipynb with live data loading, longitudinal trend visualisations (2017 to 2026), and baseline metrics.
- **Irish Freight Corridor Graph (corridor\_network.py):** Implemented NetworkX road network graph with 11 strategic nodes (Dublin, Cork, Rosslare, M50, Athlone, Limerick, Galway, Waterford, Dundalk) and 15 bidirectional corridors with tolls, speed limits, and distance matrices.
- **Baseline FIFO Dispatcher Simulator (baseline\_dispatcher.py):** Coded and executed empirical baseline simulation across 50 benchmark consignments: 8,636.5 km distance, 104.4 driving hrs, **63.5 idle waiting hrs** (averaging **1.27 hrs wasted idling per trip**), 2,879.3 L diesel, 7,601.3 kg CO2e, and €14,791.21 total cost (avg €295.82/trip).
- **Unit Test Suite:** Authored 9 unit tests across data ingestion, graph routing, and simulation; 100% passing (pytest tests -v in 1.10s).

### 2. Primary Public Dataset Sources & URLs

| Dataset Name                | URL                                                                                                                                                         | Description                                     |
|-----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------|
| CSO TBQ01 (Vessel Arrivals) | <a href="https://data.cso.ie/table/TBQ01">https://data.cso.ie/table/TBQ01</a>                                                                               | Quarterly port ship arrivals for C1 classifier  |
| CSO TBQ04 (Ro-Ro Units)     | <a href="https://data.cso.ie/table/TBQ04">https://data.cso.ie/table/TBQ04</a>                                                                               | Trailer counts (inward/outward) by port         |
| CSO TBQ05 (Trade Regions)   | <a href="https://data.cso.ie/table/TBQ05">https://data.cso.ie/table/TBQ05</a>                                                                               | Great Britain vs EU freight flows (post-Brexit) |
| EMODnet Vessel Density      | <a href="https://emodnet.ec.europa.eu/en/human-activities">https://emodnet.ec.europa.eu/en/human-activities</a>                                             | AIS coastal marine traffic density grids        |
| OpenStreetMap Ireland       | <a href="https://download.geofabrik.de/europe/ireland-and-northern-ireland.html">https://download.geofabrik.de/europe/ireland-and-northern-ireland.html</a> | National freight motorway and primary corridors |

### 3. Evaluation & Critical Reflection (So What?)

- **Validation of Research Gap:** The Week 1 baseline simulation empirically validates the core research rationale: without predictive delay awareness, freight hauliers waste an average of 1.27 hours per shipment waiting at terminal gates and customs inspections, creating direct financial and carbon penalties.
- **Data Grounding:** Live CSO PxStat API data guarantees verified, open historical maritime records (2017 to 2026) for Dublin, Cork, and Rosslare, de-risking Component 1 model development.

### 4. Action Plan & Next Steps (Now What?)

- **Oral Pitch (1 October):** Deliver the timed 1-2 minute project pitch adhering to Problem -> Gap -> Idea -> Method -> Contribution.
- **Supervisory Alignment:** Present Week 1 code, Jupyter notebook, and benchmark findings to Dr. Thanos Staikopoulos during scheduled supervisory review.
- **EMODnet & AIS Ingestion:** Incorporate EMODnet vessel density metrics into the maritime congestion feature store.

### 5. Supervisor Meeting Log & Agreed Actions

|                      |                                                                                                                                                                                               |
|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Agenda Items:        | 1. CA2 Proposal & Ethics sign-off.<br>2. 1 October Pitch rehearsal.<br>3. Demonstration of Week 1 technical deliverables: Git repo, CSO pipeline, Jupyter notebook, baseline FIFO simulation. |
| Supervisor Guidance: | - Emphasize empirical gap between unbuffered spreadsheet planning and predictive optimization.<br>- Ensure baseline simulation parameters are grounded in Irish logistics economics.          |
| Agreed Actions:      | 1. Deliver 1-2 minute pitch on 1 October.<br>2. Submit weekly progress updates every Tuesday.<br>3. Begin feature engineering on ingested CSO port traffic data.                              |

### 6. Project Milestone & Gantt Tracking

| M1 | Week 3  | ADVANCED / ON SCHEDULE | Baseline FIFO model operational; CSO data ingested; corridor graph built |
|----|---------|------------------------|--------------------------------------------------------------------------|
| M2 | Week 9  | PLANNED                | XGBoost models (C1, C2) coupled with Pareto Optimizer (C3)               |
| M3 | Week 13 | PLANNED                | Dual-system comparative evaluation & Streamlit UI dashboard              |
| M4 | Week 15 | PLANNED                | Final MSc Dissertation, code repository, and Viva Voce                   |

**Student Signature:** Rohitkumar Amritlal Jaiswal

**Date:** 29 September 2026

**Supervisor Signature:** \_\_\_\_\_

**Date:** \_\_\_\_\_